

Chapter 4

Filtering in the Frequency Domain

(overview)

2D DFT

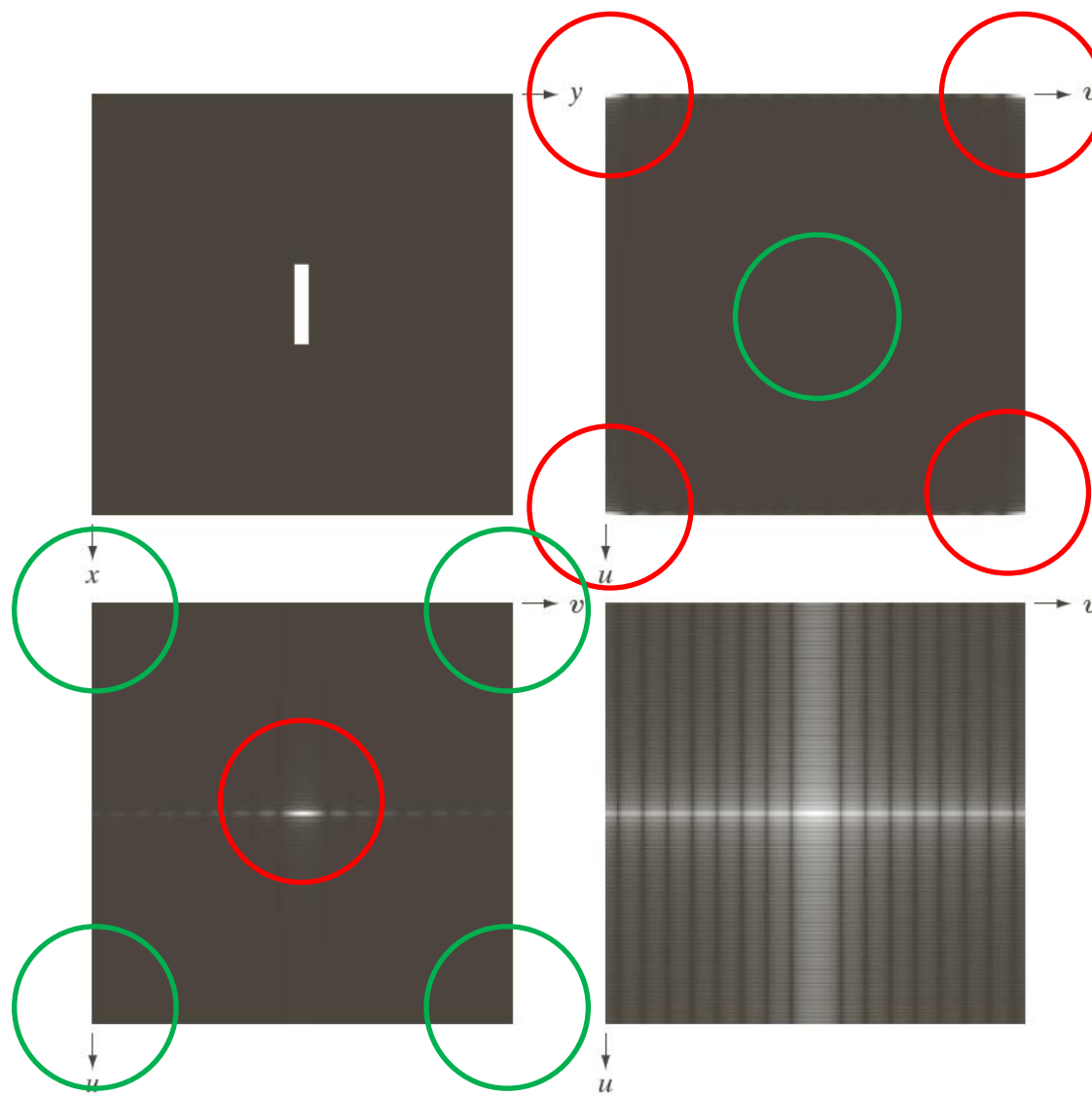
- Forward Transform

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

- Inverse Transform

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

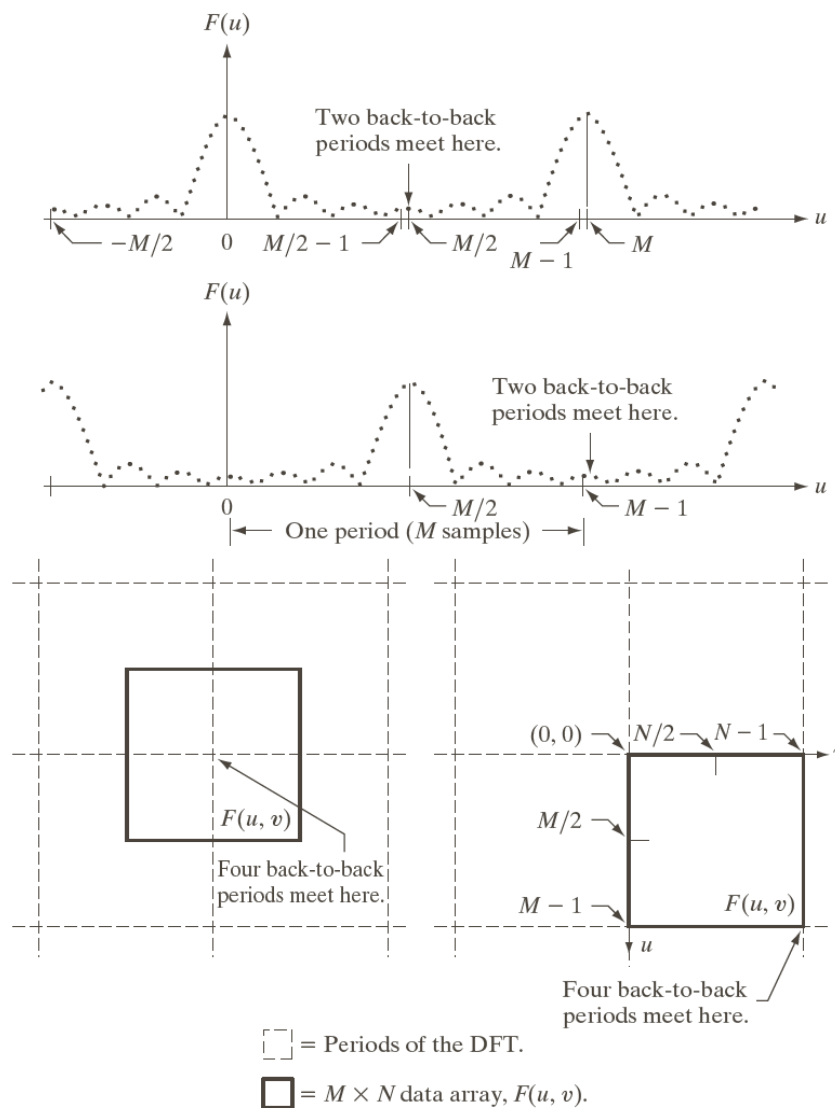
2D DFT



a	b
c	d

FIGURE 4.24

(a) Image.
(b) Spectrum showing bright spots in the four corners.
(c) Centered spectrum.
(d) Result showing increased detail after a log transformation. The zero crossings of the spectrum are closer in the vertical direction because the rectangle in (a) is longer in that direction. The coordinate convention used throughout the book places the origin of the spatial and frequency domains at the top left.



a
b
c d

FIGURE 4.23

Centering the Fourier transform.

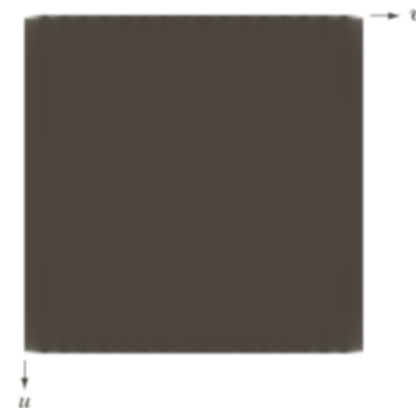
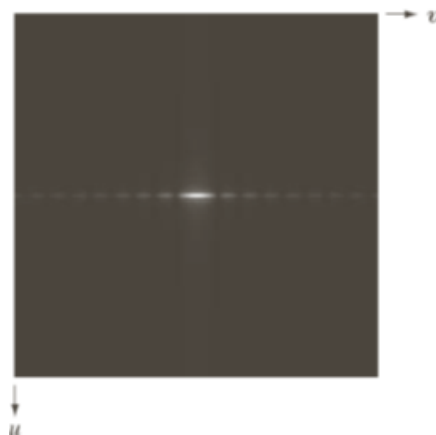
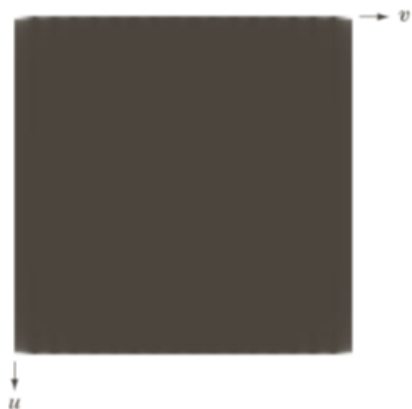
(a) A 1-D DFT showing an infinite number of periods.

(b) Shifted DFT obtained by multiplying $f(x)$ by $(-1)^x$ before computing $F(u)$.

(c) A 2-D DFT showing an infinite number of periods. The solid area is the $M \times N$ data array, $F(u, v)$, obtained with Eq. (4.5-15). This array consists of four quarter periods.

(d) A Shifted DFT obtained by multiplying $f(x, y)$ by $(-1)^{x+y}$ before computing $F(u, v)$. The data now contains one complete, centered period, as in (b).

FFT Shift(Exchange Diagonal Sub Matrix) Periodicity of DFT



Name**Expression(s)**

1) Discrete Fourier transform (DFT) of $f(x, y)$

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

3) Polar representation

$$F(u, v) = |F(u, v)| e^{j\phi(u, v)}$$

4) Spectrum

$$|F(u, v)| = [R^2(u, v) + I^2(u, v)]^{1/2}$$
$$R = \text{Real}(F); \quad I = \text{Imag}(F)$$

5) Phase angle

$$\phi(u, v) = \tan^{-1} \left[\frac{I(u, v)}{R(u, v)} \right]$$

6) Power spectrum

$$P(u, v) = |F(u, v)|^2$$

7) Average value

$$\bar{f}(x, y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$$

Name**Expression(s)**

8) Periodicity (k_1 and k_2 are integers)

$$F(u, v) = F(u + k_1M, v) = F(u, v + k_2N) \\ = F(u + k_1M, v + k_2N)$$

$$f(x, y) = f(x + k_1M, y) = f(x, y + k_2N) \\ = f(x + k_1M, y + k_2N)$$

9) Convolution

$$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n)h(x - m, y - n)$$

10) Correlation

$$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n)h(x + m, y + n)$$

11) Separability

The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.1.

12) Obtaining the inverse Fourier transform using a forward transform algorithm.

$$MNf^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v)e^{-j2\pi(ux/M+vy/N)}$$

This equation indicates that inputting $F^*(u, v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MNf^*(x, y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.2.

Separability

- How many number of multiplications and additions required to implement the 2D –DFT?

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

Separability

$$\begin{aligned} F(u, v) &= \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} = \sum_{x=0}^{M-1} e^{-j2\pi ux/M} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(vy/N)} \\ &= \sum_{x=0}^{M-1} F(x, v) e^{-j2\pi ux/M} \end{aligned}$$

$$F(x, v) = \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi vy/N}$$

Computing IDFT using DFT Algorithm (conjugate)

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

$$f^*(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v) e^{-j2\pi(ux/M + vy/N)}$$

$$MN \cdot f^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v) e^{-j2\pi(ux/M + vy/N)}$$

1D DFT & 1D FFT

$$c(n) = \frac{2^n}{n}$$

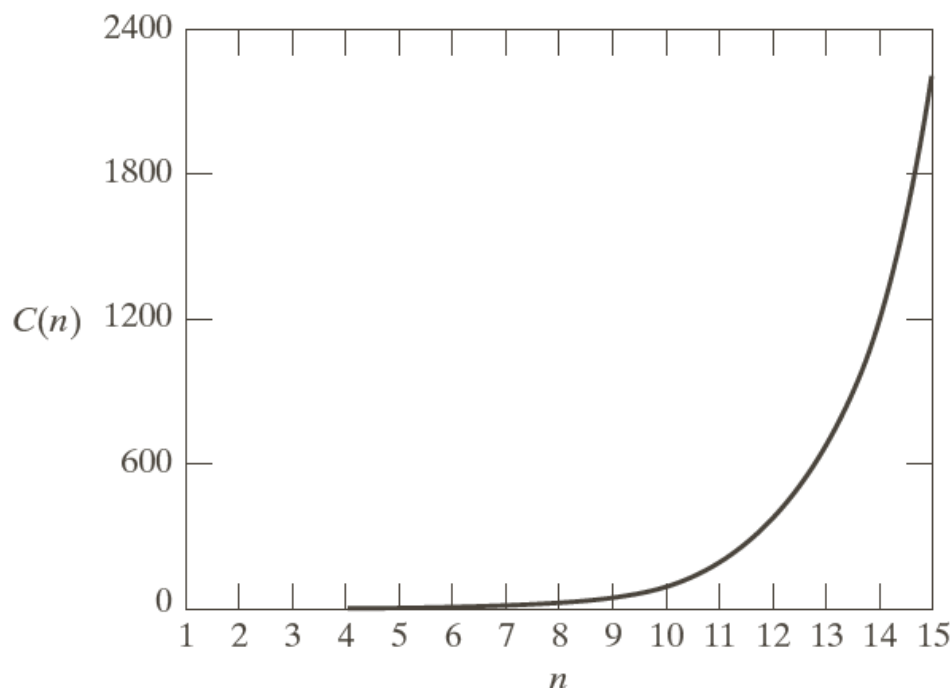
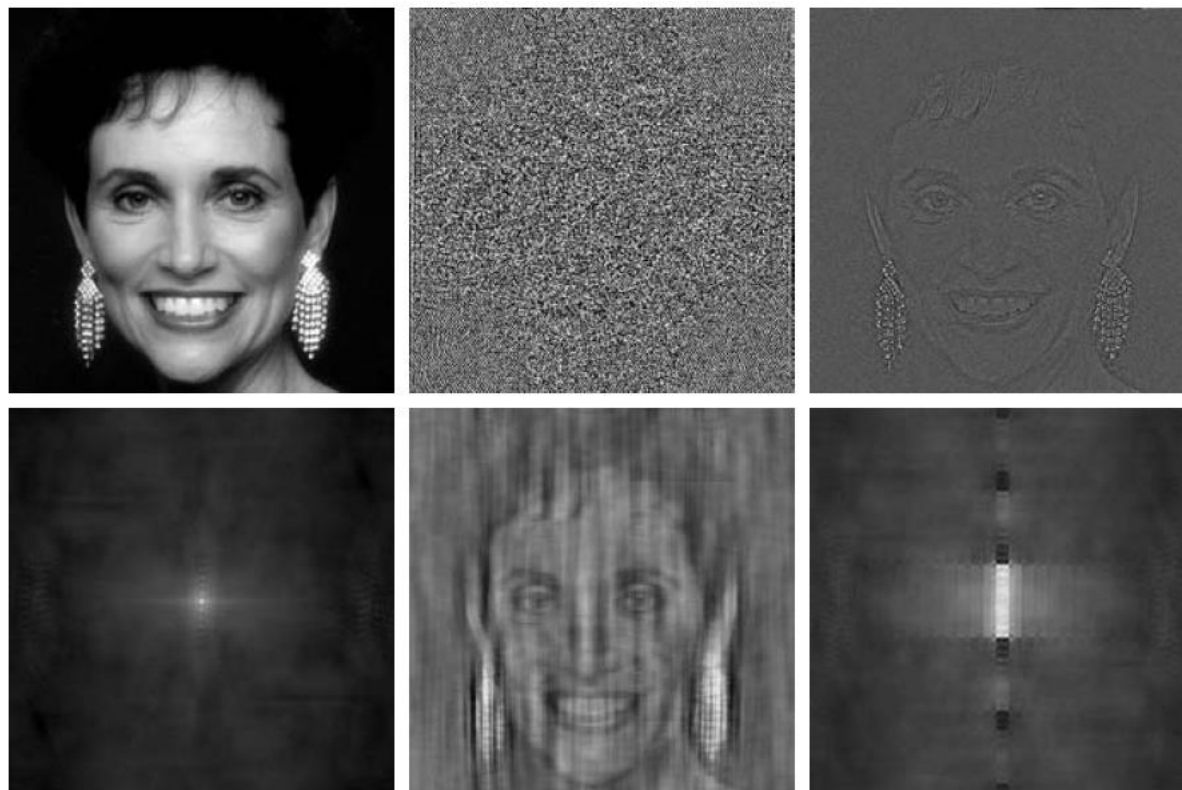


FIGURE 4.67

Computational advantage of the FFT over a direct implementation of the 1-D DFT. Note that the advantage increases rapidly as a function of n .

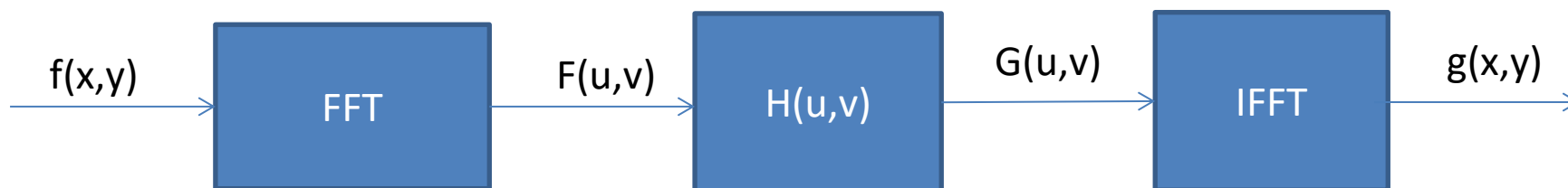
Phase Angle & Shape Information



a	b	c
d	e	f

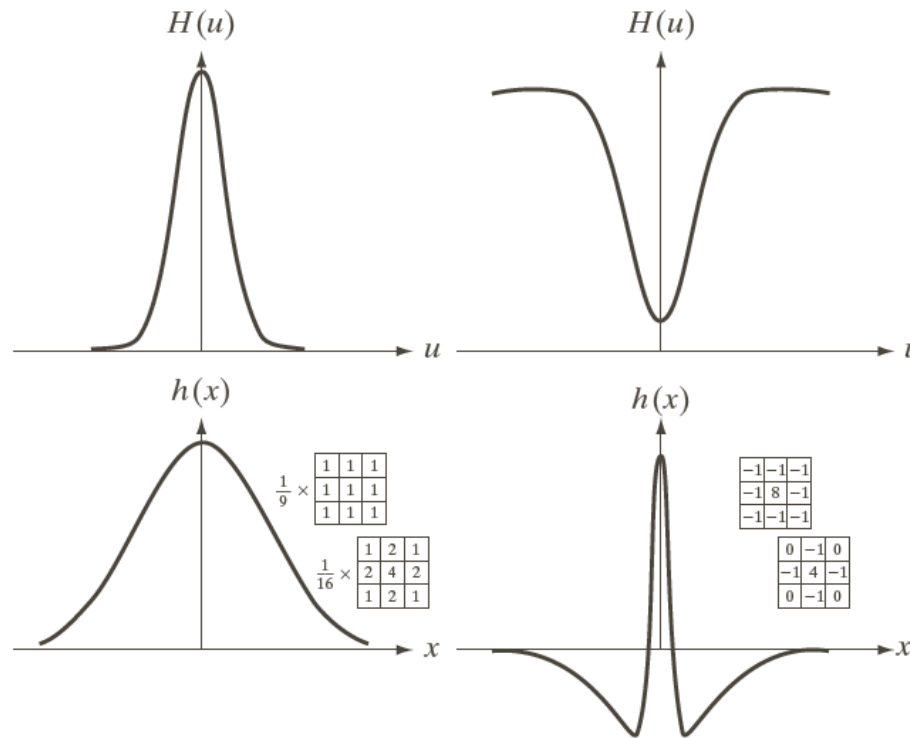
FIGURE 4.27 (a) Woman. (b) Phase angle. (c) Woman reconstructed using only the phase angle. (d) Woman reconstructed using only the spectrum. (e) Reconstruction using the phase angle corresponding to the woman and the spectrum corresponding to the rectangle in Fig. 4.24(a). (f) Reconstruction using the phase of the rectangle and the spectrum of the woman.

- $G(u,v) = F(u,v)H(u,v)$



Idea Low Pass & Ringing effect

Lowpass & Highpass Filter



a	c
b	d

FIGURE 4.37

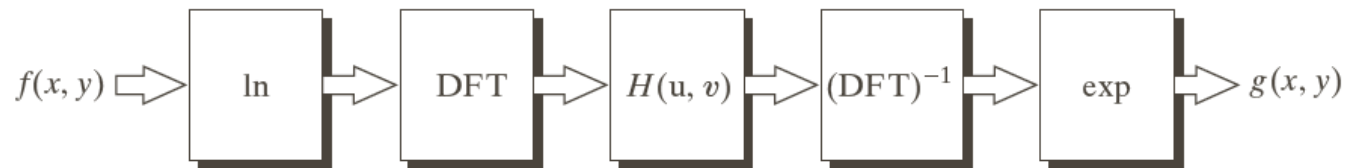
(a) A 1-D Gaussian lowpass filter in the frequency domain. (b) Spatial lowpass filter corresponding to (a). (c) Gaussian highpass filter in the frequency domain. (d) Spatial highpass filter corresponding to (c). The small 2-D masks shown are spatial filters we used in Chapter 3.

Homomorphic Filtering

- $f(x,y) = i(x,y)r(x,y)$

FIGURE 4.60

Summary of steps
in homomorphic
filtering.



Homomorphic Filtering

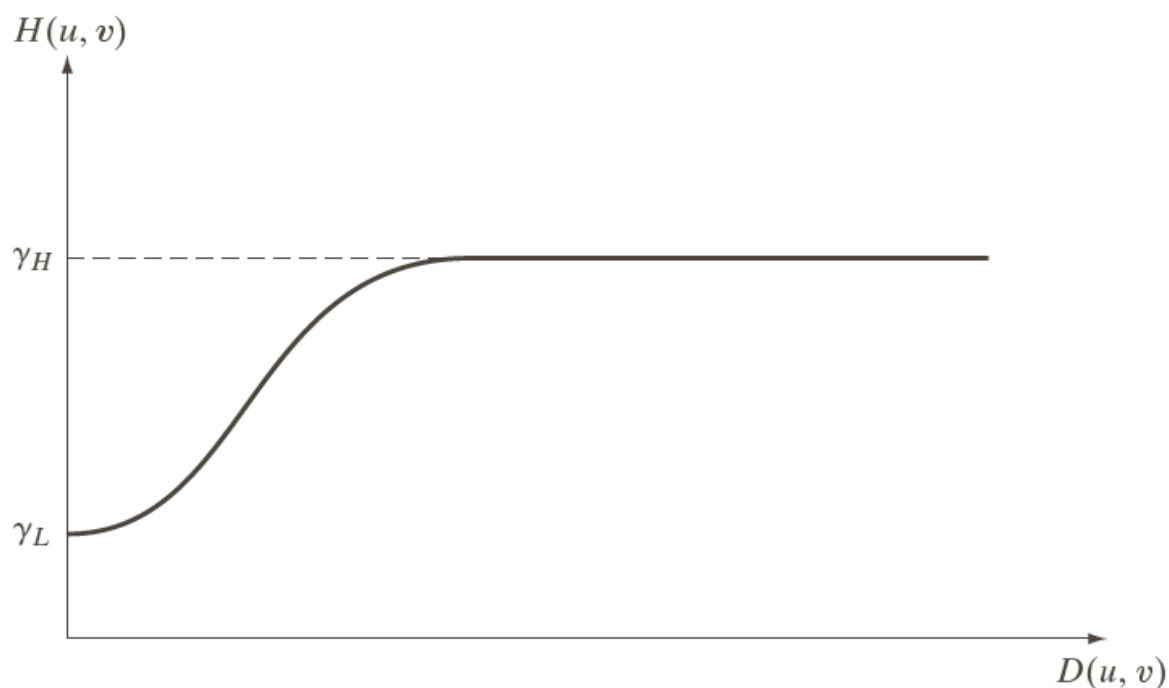
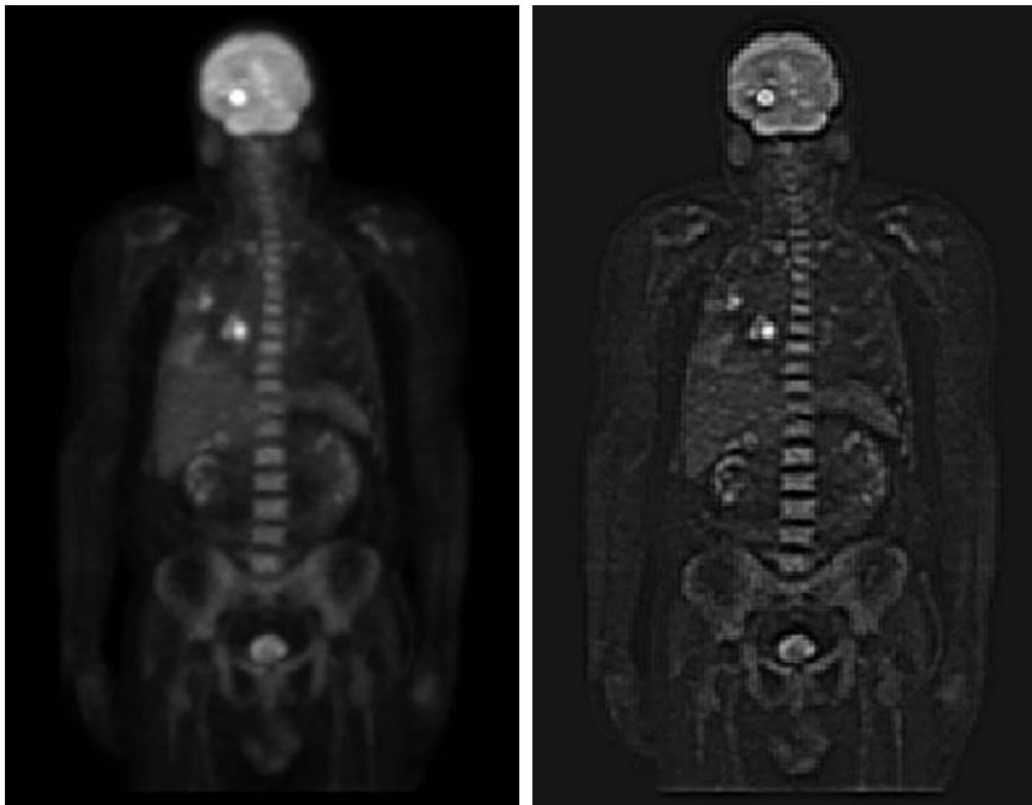


FIGURE 4.61

Radial cross section of a circularly symmetric homomorphic filter function. The vertical axis is at the center of the frequency rectangle and $D(u, v)$ is the distance from the center.

Homomorphic Filtering

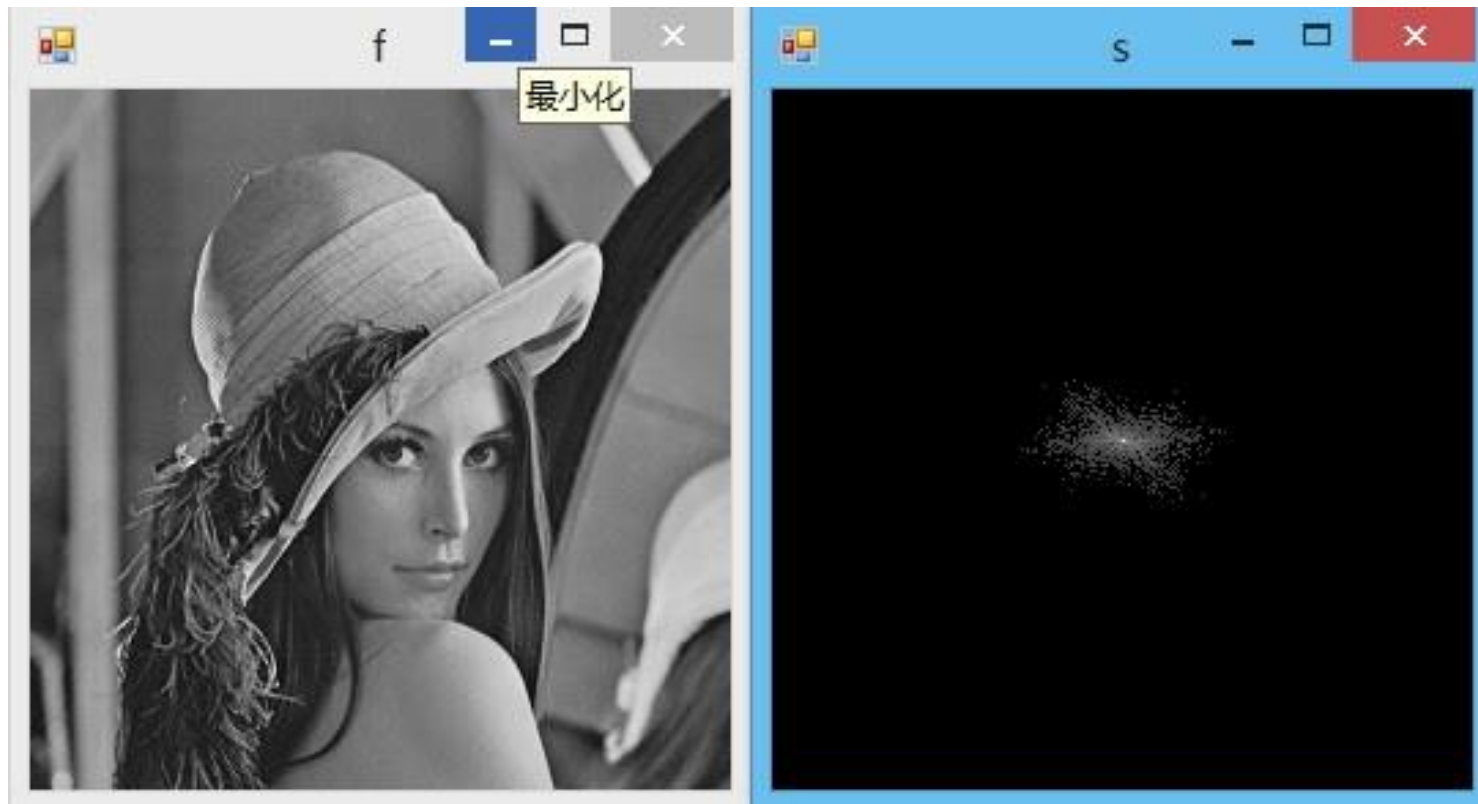


a b

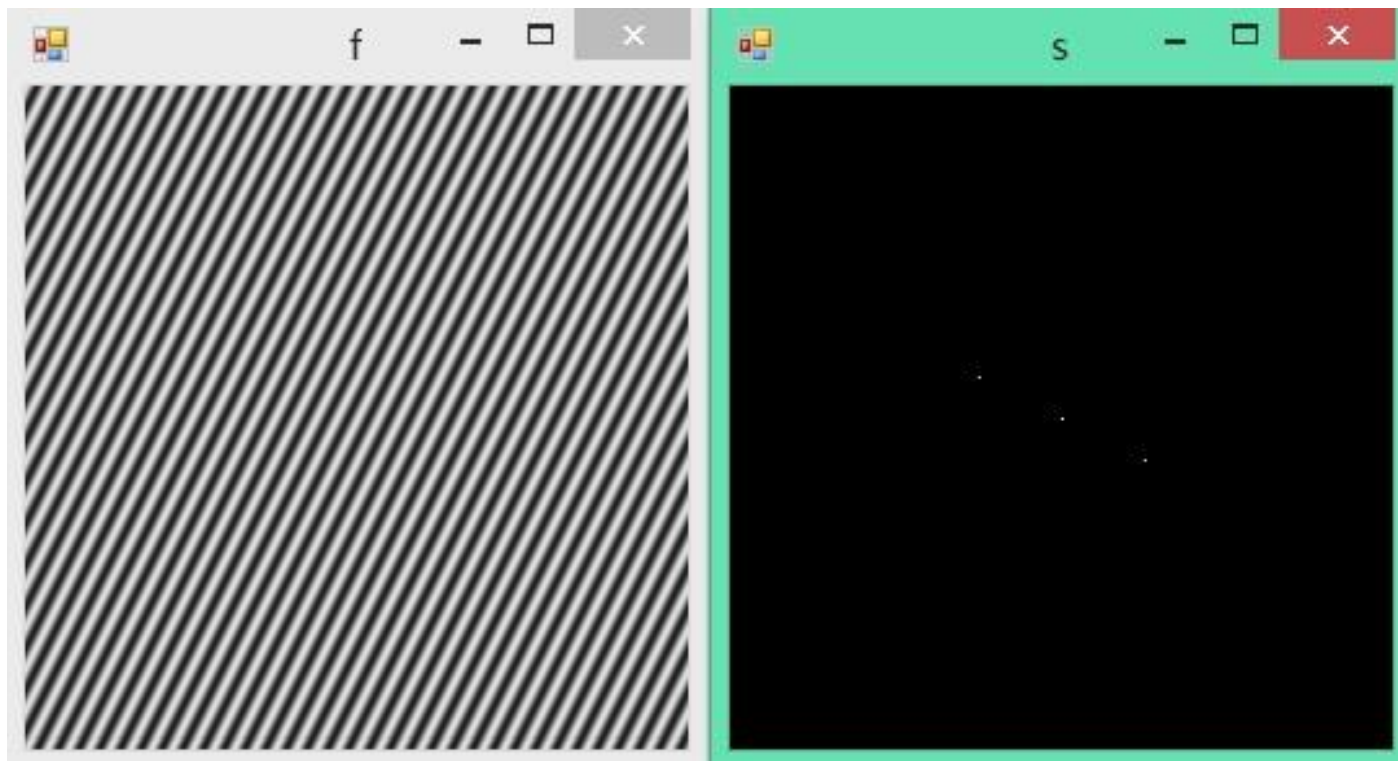
FIGURE 4.62

(a) Full body PET scan. (b) Image enhanced using homomorphic filtering. (Original image courtesy of Dr. Michael E. Casey, CTI PET Systems.)

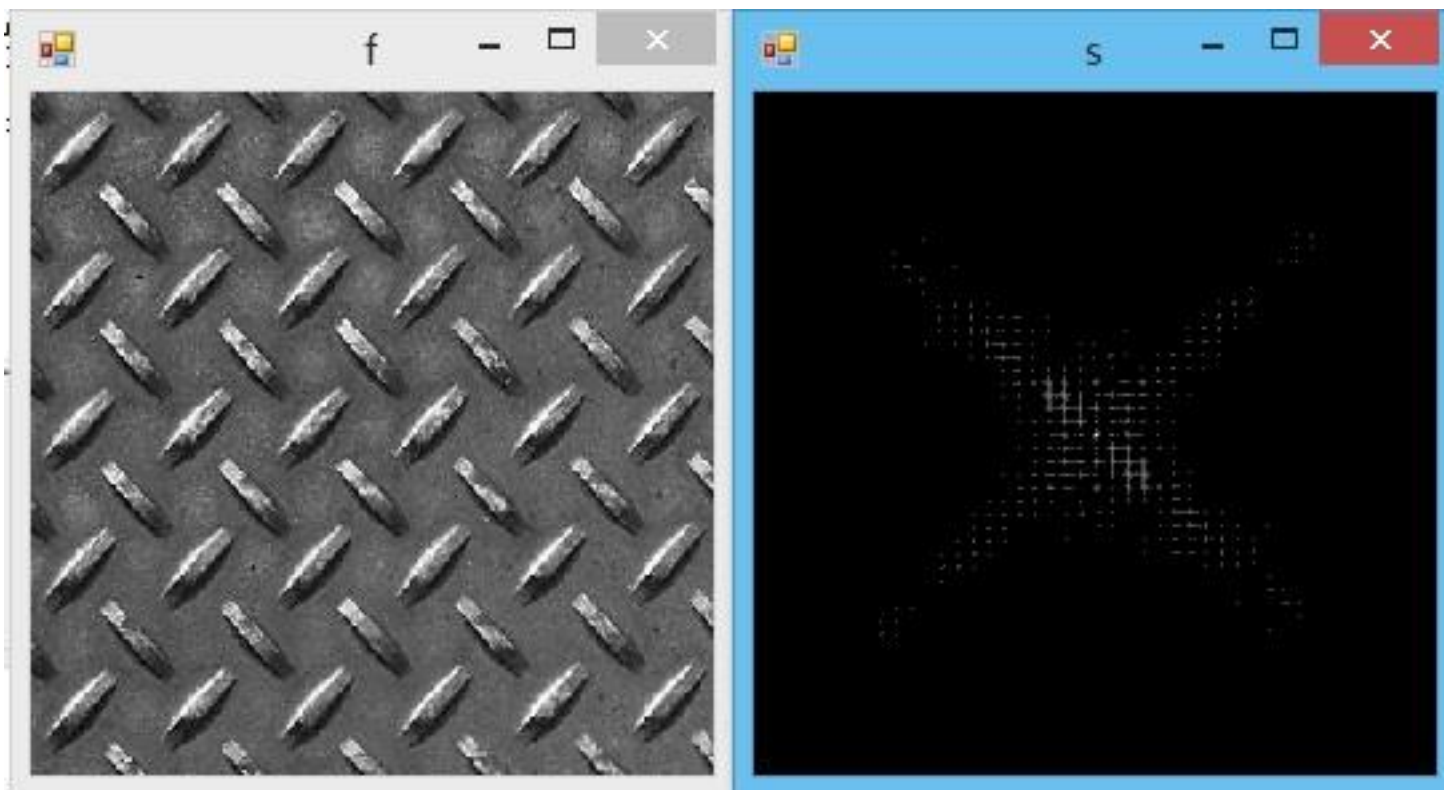
DFT of normal pictures



DFT of Sinusoidal functions



DFT of periodic images



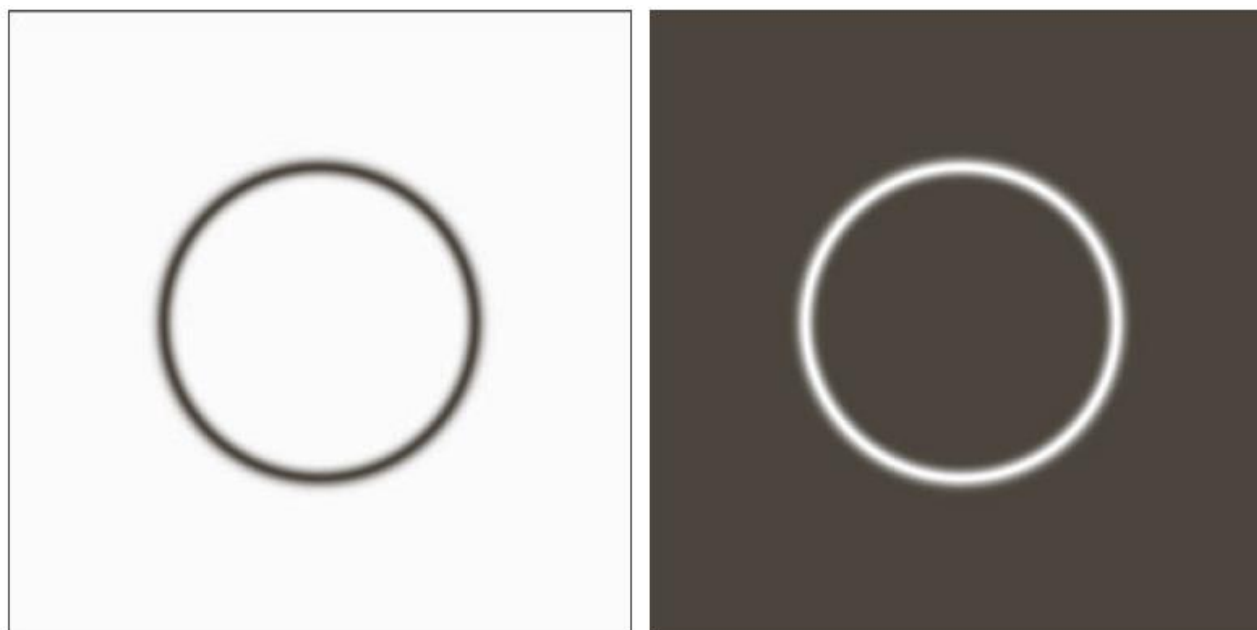


- If the signal is periodic, it can be decomposed into a combination of harmonic ones on use of Fourier series.
- 如果信号是周期性的，其可以分解成傅里叶级数使用的谐波组合。

DFT of fingerprint image



Bandreject & Bandpass



a b

FIGURE 4.63

(a) Bandreject Gaussian filter.
(b) Corresponding bandpass filter.
The thin black border in (a) was added for clarity; it is not part of the data.

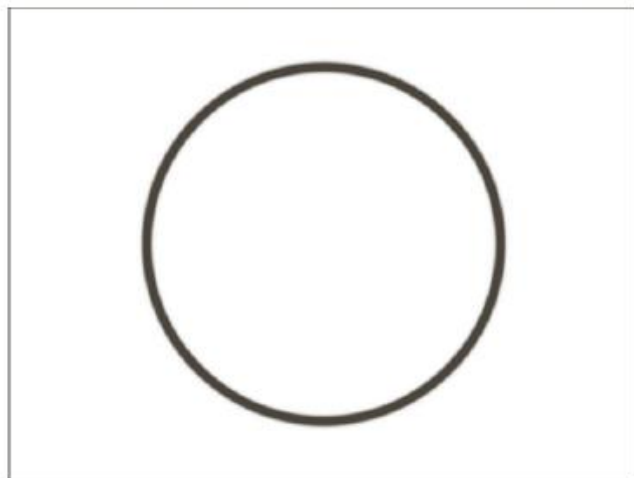
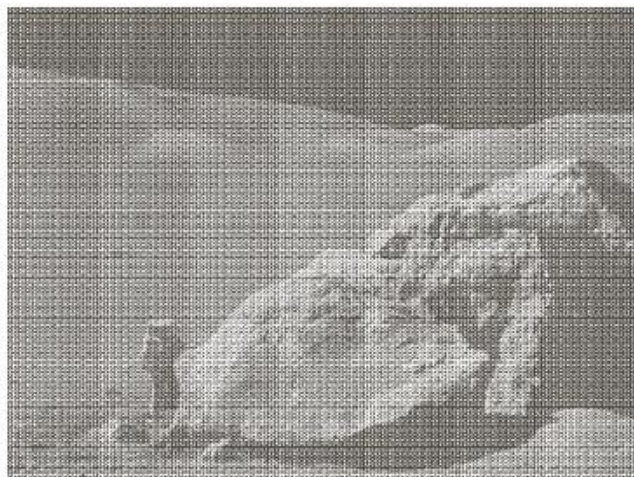
Band reject

TABLE 4.6

Bandreject filters. W is the width of the band, D is the distance $D(u, v)$ from the center of the filter, D_0 is the cutoff frequency, and n is the order of the Butterworth filter. We show D instead of $D(u, v)$ to simplify the notation in the table.

Band reject		
Ideal	Butterworth	Gaussian
$H(u, v) = \begin{cases} 0 & \text{if } D_0 - \frac{W}{2} \leq D \leq D_0 + \frac{W}{2} \\ 1 & \text{otherwise} \end{cases}$	$H(u, v) = \frac{1}{1 + \left[\frac{DW}{D^2 - D_0^2} \right]^{2n}}$	$H(u, v) = 1 - e^{-\left[\frac{D^2 - D_0^2}{DW} \right]^2}$

Band reject (sinusoidal Noise)



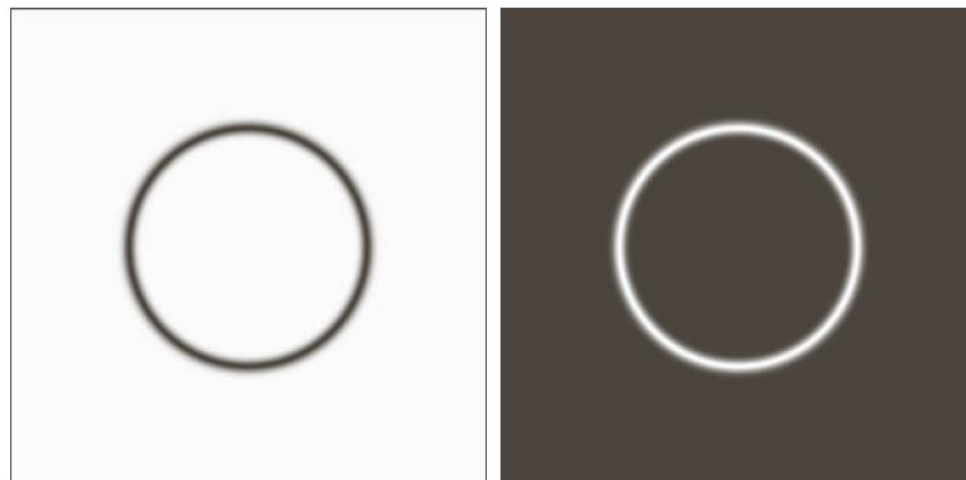
a	b
c	d

FIGURE 2.40

(a) Image corrupted by sinusoidal interference. (b) Magnitude of the Fourier transform showing the bursts of energy responsible for the interference. (c) Mask used to eliminate the energy bursts. (d) Result of computing the inverse of the modified Fourier transform. (Original image courtesy of NASA.)

Band reject

- [Programming]


TABLE 4.6

Bandreject filters. W is the width of the band, D is the distance $D(u, v)$ from the center of the filter, D_0 is the cutoff frequency, and n is the order of the Butterworth filter. We show D instead of $D(u, v)$ to simplify the notation in the table.

Ideal	Butterworth	Gaussian
$H(u, v) = \begin{cases} 0 & \text{if } D_0 - \frac{W}{2} \leq D \leq D_0 + \frac{W}{2} \\ 1 & \text{otherwise} \end{cases}$	$H(u, v) = \frac{1}{1 + \left[\frac{DW}{D^2 - D_0^2} \right]^{2n}}$	$H(u, v) = 1 - e^{-\left[\frac{D^2 - D_0^2}{DW} \right]^2}$

Notch Filter

- Notch Reject (**Demos**)

$$H_{NR}(u, v) = \prod_{k=1}^Q H_k(u, v) H_{-k}(u, v)$$

$$D_k(u, v) = [(u - M/2 - u_k)^2 + (v - N/2 - v_k)^2]^{1/2}$$

$$D_{-k}(u, v) = [(u - M/2 + u_k)^2 + (v - N/2 + v_k)^2]^{1/2}$$

- Butterworth Notch Reject

$$H_{NR}(u, v) = \prod_{k=1}^Q \left[\frac{1}{1 + [D_{0k} / D_k(u, v)]^{2n}} \right] \cdot \left[\frac{1}{1 + [D_{0k} / D_{-k}(u, v)]^{2n}} \right]$$

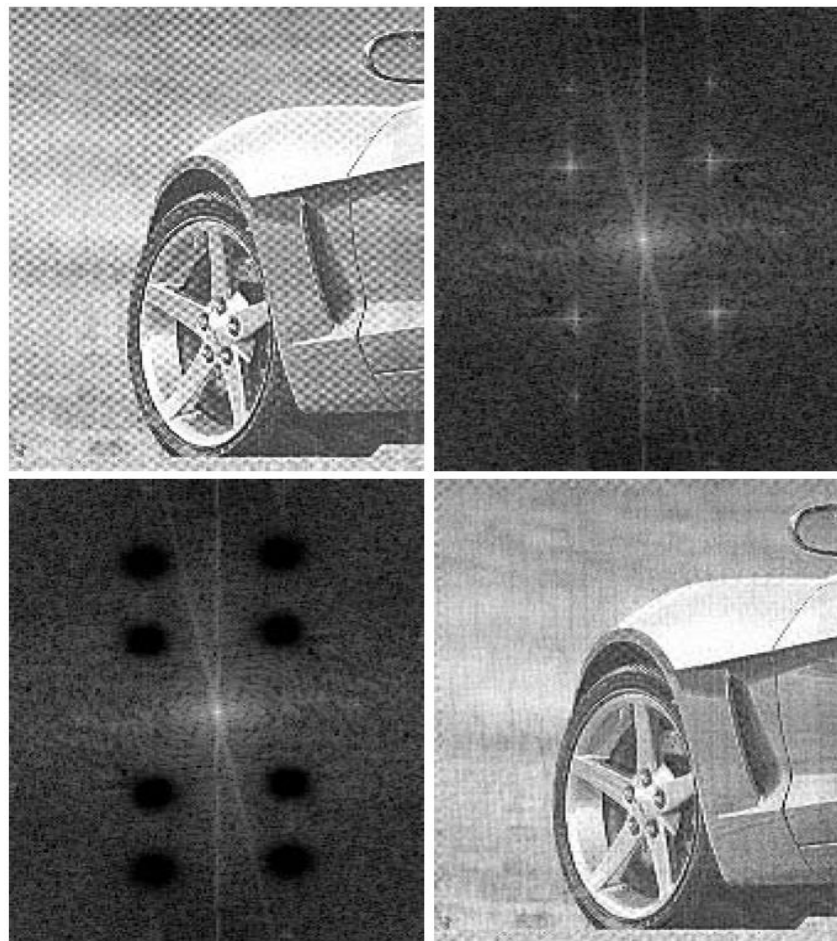
- Notch Pass $H_{NP}(u, v) = 1 - H_{NR}(u, v)$

Notch Filter

$$H_{NR}(u, v) = 1 - H_{NP}(u, v)$$

- [Discuss & Programming]

Notch reject

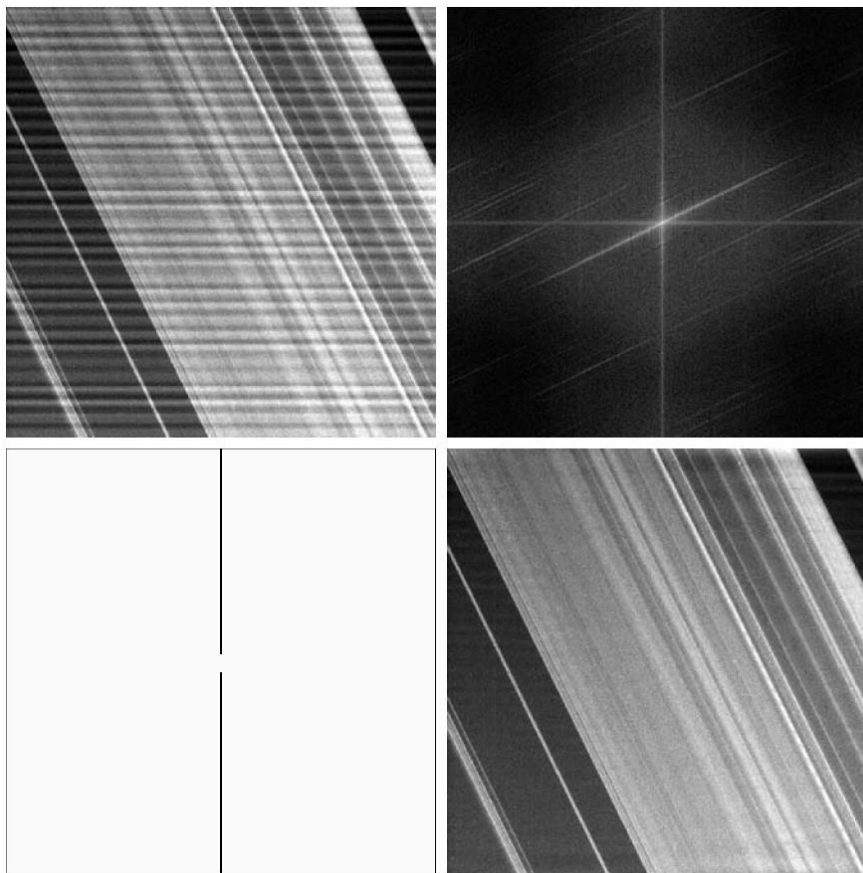


a	b
c	d

FIGURE 4.64

(a) Sampled newspaper image showing a moiré pattern.
(b) Spectrum.
(c) Butterworth notch reject filter multiplied by the Fourier transform.
(d) Filtered image.

Notch reject

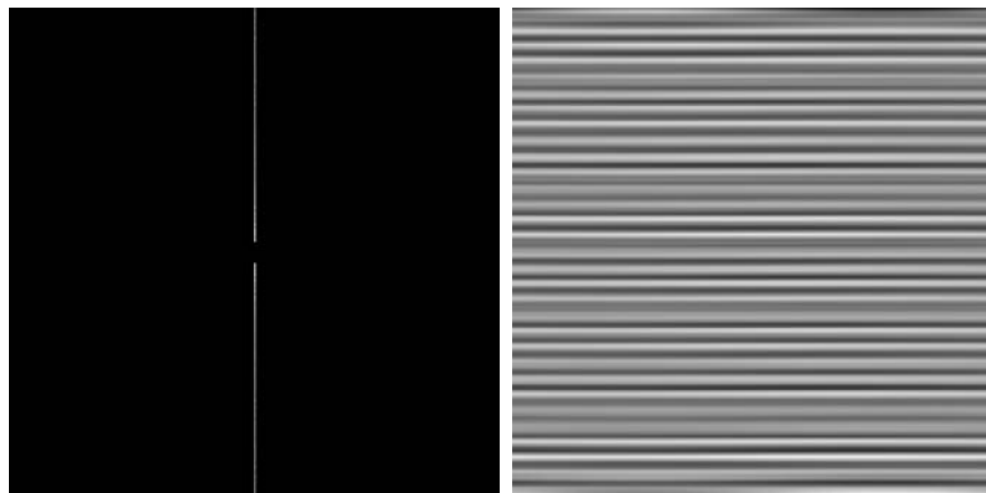


a b
c d

FIGURE 4.65

(a) 674×674 image of the Saturn rings showing nearly periodic interference. (b) Spectrum: The bursts of energy in the vertical axis near the origin correspond to the interference pattern. (c) A vertical notch reject filter. (d) Result of filtering. The thin black border in (c) was added for clarity; it is not part of the data. (Original image courtesy of Dr. Robert A. West, NASA/JPL.)

Notch Pass



a b

FIGURE 4.66

(a) Result (spectrum) of applying a notch pass filter to the DFT of Fig. 4.65(a).
(b) Spatial pattern obtained by computing the IDFT of (a).